Assignment 9.4

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Batch-13

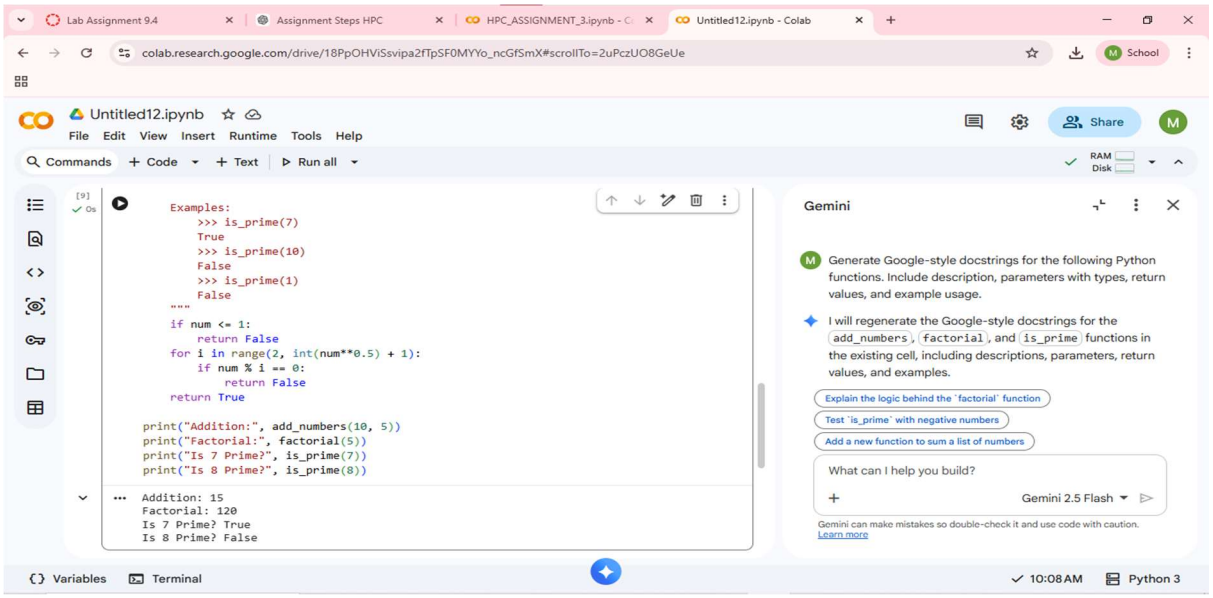
Task-1

```
def factorial(n):
    if n == 0:
        return 1
    return n * factorial(n - 1)

def is_prime(num):
    if num <= 1:
        return False
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            return False
    return True

print("Addition:", add_numbers(10, 5))
print("Factorial:", factorial(5))
print("Is 7 Prime?", is_prime(7))
print("Is 8 Prime?", is_prime(8))
```

```
Addition: 15
Factorial: 120
Is 7 Prime? True
Is 8 Prime? False
```

Prompt used:

Generate Google-style docstrings for the following Python functions. Include description, parameters with types, return value, and example usage.



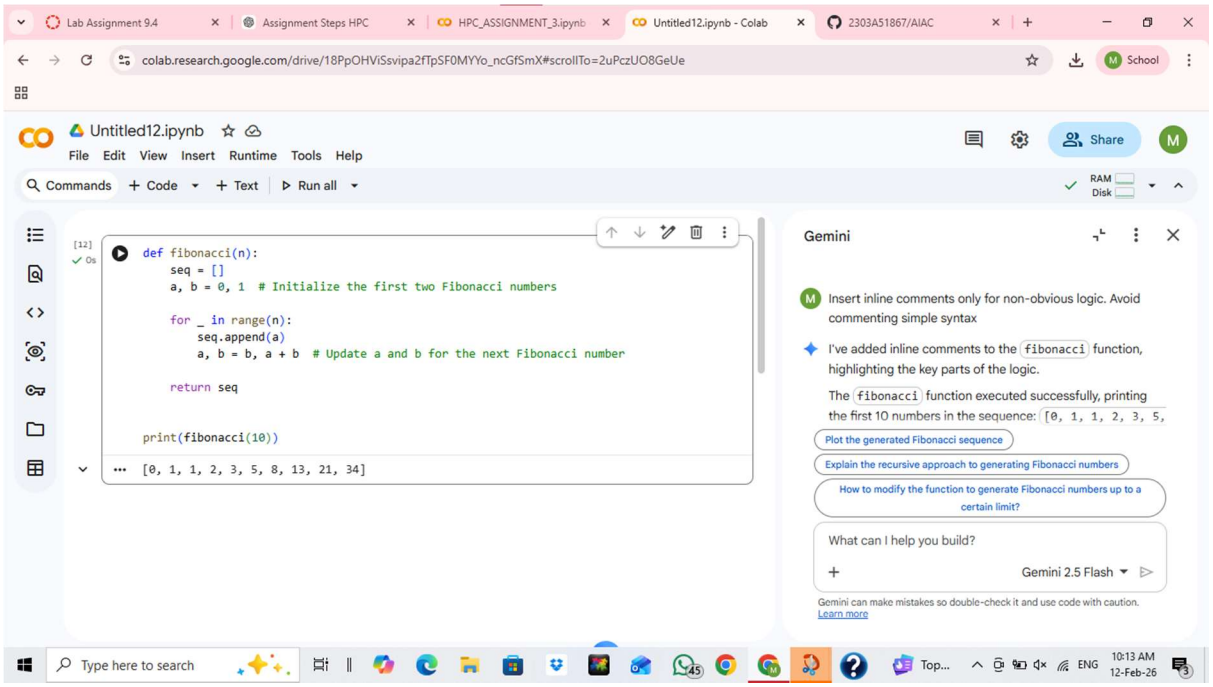
Task 2:

```
def fibonacci(n):
    seq = []
    a, b = 0, 1

    for _ in range(n):
        seq.append(a)
        a, b = b, a + b

    return seq


print(fibonacci(10))
```

```
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```

Prompt used:

Insert inline comments only for complex or non-obvious logic. Avoid commenting trivial syntax.



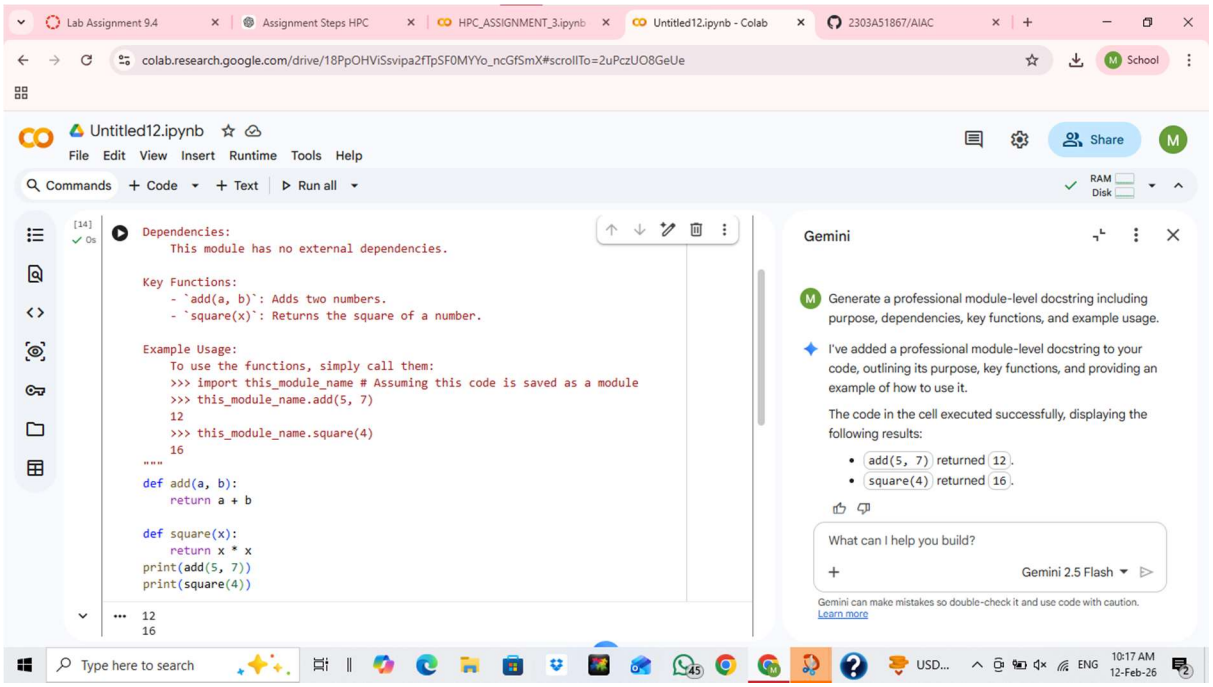
Task3:

```
def add(a, b):
    return a + b

def square(x):
    return x * x
print(add(5, 7))
print(square(4))
```

```
12
16
```

Prompt used:

Generate a professional module-level docstring including purpose, dependencies, key functions, and example usage.



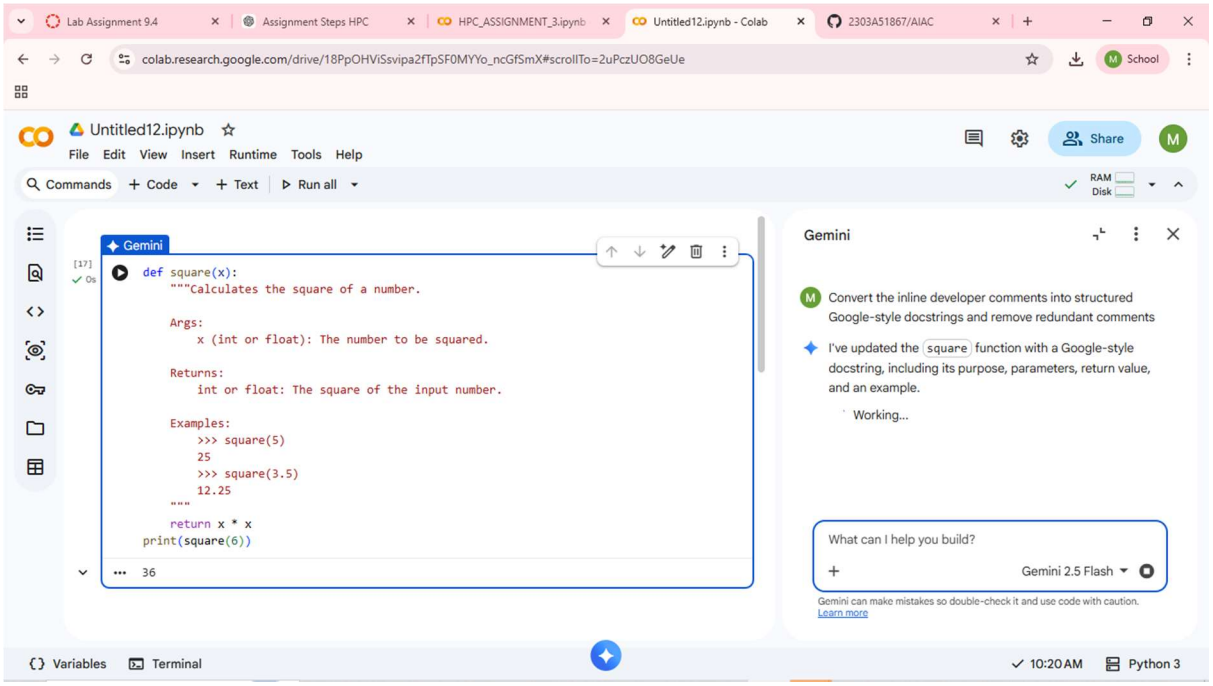
Task 4:

```
def square(x):
    return x * x
print(square(6))
```

```
36
```

Prompt used:

Convert the inline developer comments into structured Google-style docstrings and remove redundant comments.


Lab Assignment 9.4
Assignment Steps HPC
HPC_ASSIGNMENT_3.ipynb
Untitled12.ipynb - Colab
2303A51867/AIAC
colab.research.google.com/drive/18PpOHViSsvipa2fTpSF0MYYo_ncGfSmX#scrollTo=2uPczUO8GeUe
School
Untitled12.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM
Disk
Gemini
[17]
0s
def square(x):
"""Calculates the square of a number.
Args:
x (int or float): The number to be squared.
Returns:
int or float: The square of the input number.
Examples:
>>> square(5)
25
>>> square(3.5)
12.25
"""
return x * x
print(square(6))
36
Gemini
Convert the inline developer comments into structured Google-style docstrings and remove redundant comments
I've updated the square function with a Google-style docstring, including its purpose, parameters, return value, and an example.
Working...
What can I help you build?
Gemini 2.5 Flash
Gemini can make mistakes so double-check it and use code with caution.
Learn more
Variables
Terminal
10:20 AM
Python 3